

Class XI | Unit 4 | weight 10%

Operating systems

Board weight about 10% | 10 theory periods and 10 lab periods

The operating system is the program that makes the hardware usable. It starts the computer, shares the processor, keeps files, and gives you a way to command the machine.

What you should be able to do

- Define an operating system and its objectives.
- Describe batch, multitasking, time sharing, multiprocessing, parallel, distributed and embedded systems.
- Explain the main functions, including booting and process management.
- Describe process states, and distinguish a thread from a process.
- Manage files, folders, paths, search, attributes and the control panel in a GUI.
- Use the ideas behind dxdiag and msconfig, and describe open-source and mobile systems.

Path through the lecture

- 1 Define the operating system from the user's problem: hardware is difficult to command directly.
- 2 Teach features as ways of sharing a computer, using the tables.
- 3 Walk the process-state diagram with one running program and one that is waiting for a file.
- 4 Spend the lab on real files, attributes, control panel, dxdiag and msconfig.

4.1 What an operating system is

Operating system. System software that manages hardware and software resources and provides services so that the user and the application programs can use the computer.

Its objectives are convenience, efficiency and the ability to evolve. Convenience means a user can start a program, save a file and use a printer without knowing the wiring. Efficiency means the processor, memory and disks stay busy and are shared fairly. Evolution means new hardware can be added through drivers and updates.

Systems you should recognise

System	Character
DOS	Single-user, command-line operating system used on early PCs.
Windows	A widely used graphical operating system for personal computers, with multitasking.
Mac OS	The graphical operating system used on Apple personal computers.
UNIX	A multi-user, multitasking system strong on servers and on shared machines. Many later systems took ideas from it.

Ways of working

- **Batch processing** collects jobs and runs them one after another without the user sitting at the machine for each step. A payroll run is the classic example.
- **Multitasking** keeps more than one program loaded and switches the processor among them so the user seems to have several programs at once.

- **Time sharing** gives each user a short slice of processor time. The users of a shared computer each feel they have the machine.
- **Multiprocessing** uses two or more processors in one computer system.
- **Parallel processing** splits a job across multiple processors so parts of the work proceed at the same time.
- A **distributed** system spreads the work across computers that communicate over a network.
- An **embedded** operating system runs inside a device such as a washing machine, a car controller or a POS terminal. It is built for that device.

4.2 Functions

What the operating system manages

Function	What it does
Bootting	Starts the computer and loads the operating system into RAM.
Process management	Creates, schedules and removes processes.
Memory management	Allocates RAM and keeps one process out of another's memory.
File management	Names, stores, opens and protects files and folders.
I/O management	Controls input and output devices through drivers.
Secondary storage management	Tracks free space and the location of files on disks.
Network management	Provides access to network resources.
Protection	Checks users and permissions so one person cannot freely change another's work.
Command interpreter	Reads the user's command, from a typed line or from a click, and asks the system to carry it out.

Cold booting starts from a computer that was switched off. Power reaches the board, the firmware in ROM runs a power-on self-test, finds a bootable drive, and loads the operating system. **Warm booting** restarts a computer that is already on. Modern PCs use UEFI firmware for this start-up role. The curriculum name for the start-up firmware is BIOS.

4.3 Processes

Process. A program in execution. It has its own identity, its own state, and its own address space.

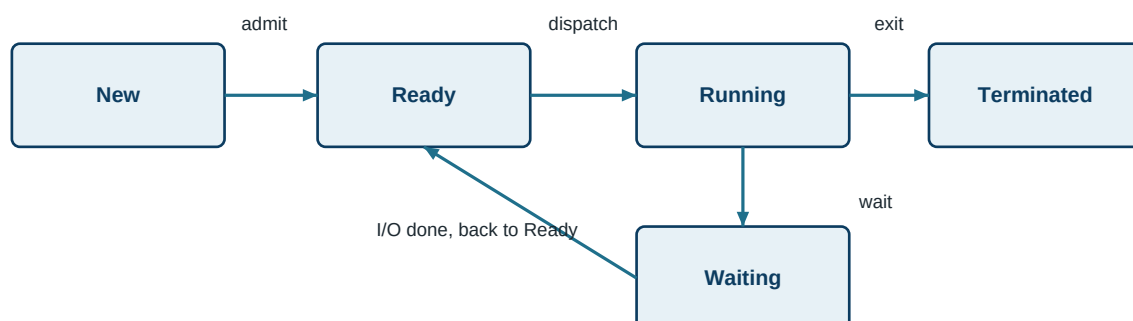


Figure 4.1 States of a process.

- **New:** the process is being created.
- **Ready:** the process has everything it needs except the processor.

- **Running:** the processor is executing its instructions.
- **Waiting or blocked:** the process cannot continue until an event finishes, often an input or output operation.
- **Terminated:** execution is finished and the operating system reclaims the resources.

A **thread** is a path of execution inside a process. Threads of one process share that process's memory. A **process** has its own memory space. **Multithreading** means one process has several threads, for example one thread drawing a page while another loads an image. **Multitasking** means the system switches among several processes. Multiprocessing means more than one processor is present.

4.4 Working in a graphical operating system

Windows presents a desktop, icons, a taskbar and windows. **Windows Explorer**, now called File Explorer, is the program that shows drives, folders and files. You create a folder, delete it, copy it, rename it, and move it by cut and paste or by dragging. A copy leaves the original in place. A move takes it to the new place.

A **drive** is a storage device with a letter, such as C. A **path** is the address of a file. An absolute path starts from the drive, for example `C:\College\CS\notes.txt`. A relative path starts from the folder you are already in. The backslash separates folder names in Windows.

Search finds a file by name, by part of a name, or by date. **Attributes** are flags stored with the file. **Read-only** refuses an ordinary save. **Hidden** keeps the file out of the ordinary listing. **System** marks a file the operating system depends on. **Archive** marks a file that should be included in the next backup. You can inspect and change attributes from the file's properties.

The **control panel** is the place where hardware and software are adjusted: display, sound, user accounts, date and time, devices, and programs. `dxdia` opens the DirectX diagnostic tool, which reports the computer's system, display and sound resources. `msconfig` opens System Configuration, which shows start-up choices and services. Change `msconfig` only when you understand the item, because a wrong boot setting can stop the machine starting normally.

4.5 Other operating systems

Linux is the usual classroom example of an operating system other than Windows. **Open-source** software is distributed with its source code. People may study it, share it, and modify it under the licence. The advantages are a low licence cost, the right to inspect the code, a community that repairs faults in public, and the freedom to adapt the system.

Android is an open-source operating system for mobile devices, built on a Linux kernel. Phones show a version number in the settings. Older classroom charts also use dessert codenames such as KitKat and Lollipop. What you need to write is that Android is a mobile operating system, it is versioned, and each version adds features and repairs.

In the lab

In the lab, create a folder CS-XI on the desktop, copy a file into it, rename the copy, make it read-only, hide it, then search for it by name. Open `dxdia` and read the operating system name and the memory size. Open `msconfig`, look at the Services tab, and close it without saving a change.

Key terms

Term	Meaning to write in an answer
Operating system	Software that manages resources and provides a way to use the computer.
Bootling	The start-up sequence that loads the operating system.

Term	Meaning to write in an answer
Process	A program that is being executed.
Thread	A single path of execution inside a process.
Multitasking	Sharing one processor among several tasks by switching.
Time sharing	Giving each user a slice of processor time.
Path	The location of a file, written as drive and folders.
Open source	Software whose source code may be studied and modified under its licence.

Bring it together

- The operating system makes hardware usable and shares it safely.
- Batch, interactive multitasking, time sharing, multiprocessing, parallel, distributed and embedded are different ways of organising that work.
- A process moves through new, ready, running, waiting and terminated.
- Files live at paths and carry attributes. The control panel, dxdiag and msconfig are the Windows tools named in this unit.

Do this in the practical

- 1 Show the desktop, File Explorer, a drive and a path.
- 2 Create, rename, copy, move and delete a folder. Search for the file afterwards.
- 3 Set and clear the read-only and hidden attributes.
- 4 Open control panel, dxdiag and msconfig. Record the operating system name and the installed RAM. Do not save a change in msconfig.

Check yourself

Attempt every item before you read the key. Section A of the Computer Science paper is a set of multiple-choice questions.

1. A program in execution is a:

- A. File attribute
- B. Process
- C. Port
- D. Nibble

2. Which state means the process is waiting for the processor?

- A. Terminated
- B. Ready
- C. New only
- D. ROM

3. Time sharing is used so that:

- A. Each user gets a slice of CPU time
- B. Files are compressed
- C. ROM is erased
- D. A plotter draws faster

4. An embedded OS is found in:

- A. A device built for a specific product
- B. A compiler only
- C. A nibble
- D. A barcode definition

5. The hidden attribute:

- A. Speeds up the ALU
- B. Keeps a file out of the ordinary listing
- C. Converts AC to DC
- D. Is a network topology

6. dxdiag is used to:

- A. Diagnose DirectX system, display and sound information
- B. Erase EPROM with ultraviolet light
- C. Design a bus topology
- D. Compile C++

7. Threads of the same process:

- A. Must each have a separate hard disk
- B. Share that process's memory
- C. Are always stored in ROM
- D. Replace the power supply

8. Open-source software allows the user to:

- A. Study and modify the source under the licence
- B. Remove the need for a processor
- C. Disable every attribute
- D. Turn DRAM into a printer

Answer key

Q	Answer	Why this is the answer
1	B	A process is a program that is running.
2	B	A ready process can run as soon as the scheduler dispatches it.
3	A	Time sharing switches the CPU among users quickly.
4	A	Embedded systems run inside dedicated devices.
5	B	Hidden is a file attribute that suppresses the normal display.
6	A	dxdiag reports the multimedia and system resources Windows can see.
7	B	Threads are lightweight paths of execution inside one process address space.
8	A	The source is available for study and modification under its licence.

Short answers

A 3-mark answer needs a definition and two clear points, or a three-row difference. Write in sentences.

Q1. Define an operating system.

Model answer

An operating system is system software that manages hardware and software resources and provides services through which the user and application programs use the computer.

Q2. Differentiate multitasking and multiprocessing.

Model answer

Multitasking switches one processor among several tasks so they appear to run together. Multiprocessing uses two or more processors in the computer. A machine can do both: several processors, each shared among tasks.

Q3. Differentiate a process and a thread.

Model answer

A process is a running program with its own address space. A thread is one path of execution inside a process. Threads of a process share its memory. Creating a thread costs less than creating another process.

Q4. List the five states of a process.

Model answer

New, ready, running, waiting (blocked), and terminated. A new process is admitted to ready. The scheduler dispatches a ready process to running. It may wait for input or output, return to ready when that finishes, or terminate.

Q5. What is a file path? Give one example.

Model answer

A path is the location of a file in the folder tree. An absolute Windows path includes the drive, for example C:\College\CS\notes.txt.

Detailed answers

Section C asks for a connected answer. Open with a definition, develop the points below, and close with one example.

Question 1. Explain the main functions of an operating system.

- 1 Define the operating system and its objectives: convenience, efficiency, evolution.
- 2 Explain booting, process management, memory management and file management.
- 3 Explain I/O, secondary storage, network, protection and the command interpreter.
- 4 Give one example for protection, such as a user account and a read-only file.

Question 2. Describe process states and distinguish multithreading from multitasking.

- 1 Define a process.
- 2 Draw or describe new, ready, running, waiting and terminated, with the reason for each move.
- 3 Define a thread and say that threads share a process's memory.
- 4 Multithreading is many threads in one process. Multitasking is many processes sharing the processor. Multiprocessing is many processors.